

Seunghyun Shin

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EDUCATION

Mar. 2021 ~	Gwangju Institute of Science and Technology Artificial Intelligence Graduate School <i>M.S/Ph.D. Integrated Program (Ph.D.)</i>	Gwangju, Republic of Korea
Mar. 2017 ~ Feb. 2021	Gwangju Institute of Science and Technology Department of Electrical Engineering and Computer Science <i>B.S in Electronic Engineering and Computer Science</i>	Gwangju, Republic of Korea

RESEARCH EXPERIENCE

Sep. 2025 ~ Mar. 2026	Research Intern at Huawei Noah's Ark Lab <i>Camera Team in London Office</i>	London, United of Kingdom
Jun. 2023 ~ Aug. 2023	Research Intern at University of California San Diego <i>Video Processing Lab in Department of Electrical and Computer Engineering</i>	California, United of States
Mar. 2020 ~ Feb. 2021	Research Intern at Gwangju Institute of Science and Technology <i>Visual AI Lab in Department of Electronic Engineering and Computer Science</i>	Gwangju, Republic of Korea

RESEARCH INTEREST

- Generative Model – Video Generation, Video Editing, Kinety Typography Generation
- Creative AI – Aesthetic Image / Video Enhancement, Video Color Grading, Image Decolorization
- Computational Photography – Image Super Resolution, Denoising, Low-Light Image Enhancement

PUBLICATIONS

1. Wooseok Jeon*, Seungho Park*, **Seunghyun Shin**, Sangeyl Lee, Hyeonho Jeon and Hae-Gon Jeon, “Rebalancing Reference Frame Dominance to Improve Motion in Image-to-Video Models” *In Proc. of the 36th International Conference on Neural Information Processing Systems (NeurIPS)*, Dec 2026
2. **Seunghyun Shin**, Jifei Song, Wooseok Jeon, Hae-Gon Jeon and Jiankang Deng, “TriMotion: Modality-Agnostic Camera Control for Video Generation” *In Proc. of European Conferences on Computer Vision (ECCV)*, Sep 2026
3. Seonmi Park, **Seunghyun Shin**, Vihaan Misra, Dongmin Shin, Ukcheol Shin, Jean Oh and Hae-Gon Jeon, “Bridging Online and Offline Handwriting via Differentiable Physical Rendering” *In Proc. of European Conferences on Computer Vision (ECCV)*, Sep 2026
4. Chanhui Lee, Donggyu Choi, **Seunghyun Shin**, Hae-Gon Jeon and Jeany Son, “Universal Image Immunization against Diffusion-based Image Editing via Semantic Injection” *In Proc. of European Conferences on Computer Vision (ECCV)*, Sep 2026
5. Sang-Hun Han, Min-Gyu Park, Jisu Shin, **Seunghyun Shin**, Jin-Hwi Park and Hae-Gon Jeon, “PIAvatar: Physically Interactive Avatars via Deformation Gradient Decoupling” *In Proc. of European Conferences on Computer Vision (ECCV)*, Sep 2026
6. Wooseok Jeon, **Seunghyun Shin**, Dongmin Shin and Hae-Gon Jeon, “Motion Prior Distillation in Time Reversal Sampling for Generative Inbetweening” *In Proc. of the International Conference on Learning Representations (ICLR)*, May 2026
7. Jisu Shin, Richard Shaw, **Seunghyun Shin**, Zhensong Zhang, Hae-Gon Jeon and Eduardo Pérez-Pellitero, “CHROMA: Consistent Harmonization of Multi-View Appearance via Bilateral Grid Prediction” *In Proc. of the International Conference on Learning Representations (ICLR)*, May 2026
8. **Seunghyun Shin**, Dongmin Shin, Jisu Shin, Hae-Gon Jeon and Joon-Young Lee, “Video Color Grading via Look-Up Table Generation” *In Proc. of the IEEE International Conference on Computer Vision (ICCV)*, Oct 2025

9. Seonmi Park, Inhwon Bae, **Seunghyun Shin** and Hae-Gon Jeon, “Kinetic Typography Diffusion Model.” *In Proc. of European Conferences on Computer Vision (ECCV)*, Oct 2024
10. **Seunghyun Shin**, Jisu Shin, Jihwan Bae, Inwook Shim and Hae-Gon Jeon, “Close Imitation of Expert Retouching for Black-and-White Photography” *In Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun 2024
11. **Seunghyun Shin**^{*}, Jisu Shin^{*} and Hae-Gon Jeon, “Task-specific Scene Structure Representations.” *In Proc. of Association for the Advancement of Artificial Intelligence (AAAI)*, Feb 2023, [Oral Presentation]

HONORS & AWARDS

- Outstanding Reviewer (Top 3%), European Conferences on Computer Vision (ECCV’26), Jul 2026
- Doctoral Consortium, European Conferences on Computer Vision (ECCV’26), Jul 2026
- Finalist of Qualcomm Innovation Fellowship Korea, Qualcomm Technologies, Inc., Nov 2024.
- Program Committee for The Association for the Advancement of Artificial Intelligence (AAAI’25), Aug 2024
- International Computer Vision Summer School, Sicily, Italy, Jul 2024.
- Overseas Internship Program Scholarship Funding, AI Graduate School, GIST, March 2023.
- Excellent Paper Award, The 29th International Workshop on Frontiers of Computer Vision, Feb 2023.
- RA Student Research Achievement Scholarship, AI Graduate School, GIST, Dec 2021, 2022, 2023, 2024.
- Best B.S. Thesis Award, GIST, Feb 2021
- Selected as Summer School Exchange Program Scholarship on UC Berkeley, Jun 2018 ~ Aug 2018

ACADEMIC SERVICES

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) - 2024, 2025, 2026
- The Association for the Advancement of Artificial Intelligence (AAAI) - 2025
- IEEE International Conference on Computer Vision (ICCV) - 2025
- Advances in Neural Information Processing Systems (NeurIPS) – 2025, 2026
- International Conference on Learning Representations (ICLR) – 2026, 2027
- European Conference on Computer Vision (ECCV) – 2026
- The British Machine Vision Conference (BMVC) – 2026
- Neural Networks, Elsevier

PATENTS

- H.G. JEON, S.H. HAN, **S.H. SHIN**, J.S. SHIN, J.H. PARK, “METHOD AND APPARATUS FOR GENERATING MOTION OF AVATAR”, *Korea Patent App. 10-2026-0073831*
- W.S. JEON, **S.H. SHIN**, D.M. SHIN, H.G. JEON, “MOTION PRIOR DISTILLATION IN TIME REVERSAL SAMPLING FOR GENERATIVE INBETWEENING”, *Korea Patent App. 10-2026-0073831*
- **S.H. SHIN**, D.M. SHIN, J.S. SHIN, H.G. JEON, “METHOD AND APPARATUS FOR GENERATING ADAPTIVE LOOK-UP TABLE FOR VIDEO COLOR GRADING BASED ON HIGH-DIMENSIONAL COLOR FEATURES USING DIFFUSION MODEL”, *Korea Patent App. 10-2026-0059237*
- S.M. PARK, I.H. BAE, **S.H. SHIN**, H.G. JEON, “KINETIC TYPOGRAPHY VIDEO GENERATION SYSTEM AND CONTROL METHOD THEREOF, AND LEARNING METHOD OF KINETIC TYPOGRAPHY VIDEO GENERATION SYSTEM”, *Korea Patent App. 10-2026-0059237 / US Patent App. 19/366,853*
- **S.H. SHIN**, J.S. SHIN, J.H. BAE, I.W. SHIM, H.G. JEON, “METHOD FOR CREATING BLACK-AND-WHITE PHOTO THAT IMITATES EXPERT RETOUCHING STYLES”, *Korea Patent App. 10-2023-0030667 / US Patent No. 12,620,071*
- **S.H. SHIN**, J.S. SHIN, H.G. JEON, “METHOD FOR GENERATING TASK-SPECIFIC SCENE STRUCTURE”, *Korea Patent No. 10-2720637 / US Patent App. 18/508,456*
- **S.H. SHIN**, J.S. SHIN, H.G. JEON, “IMAGE HARMONIZATION METHOD THROUGH INPAINTING BOUNDARIES OF FOREGROUND IMAGE”, *Korea Patent No. 10-2562677 / US Patent App. 18/410,525*

^{*} These authors contributed equally.